

Covariance matrix Evolution and Lyapunov spectra

The many-body density evolves as follows

Define the stochastic quantum state maps

$$\text{filling: } \tilde{\mathcal{E}}_{\vec{r},-,v}(\hat{\rho}) = \xi_{\vec{r},-,v} \frac{\hat{N}_{\vec{r},-,v} \hat{\rho} \hat{N}_{\vec{r},-,v}}{\text{Tr}(\hat{\rho} \hat{N}_{\vec{r},-,v})} + (1 - \xi_{\vec{r},-,v}) \frac{\hat{X}_{\vec{r},-,v}^\dagger \hat{\rho} \hat{X}_{\vec{r},-,v}}{\text{Tr}(\hat{\rho} (1 - \hat{N}_{\vec{r},-,v}))}$$

$$\text{depletion: } \tilde{\mathcal{E}}_{\vec{r},+,v}(\hat{\rho}) = \xi_{\vec{r},+,v} \frac{\hat{X}_{\vec{r},+,v} \hat{\rho} \hat{X}_{\vec{r},+,v}^\dagger}{\text{Tr}(\hat{\rho} \hat{N}_{\vec{r},+,v})} + (1 - \xi_{\vec{r},+,v}) \frac{(1 - \hat{N}_{\vec{r},+,v}) \hat{\rho} (1 - \hat{N}_{\vec{r},+,v})}{\text{Tr}(\hat{\rho} (1 - \hat{N}_{\vec{r},+,v}))}$$

$$\text{Here, } \xi_{\vec{r},n,v} \sim \text{Ber}(\langle \hat{N}_{\vec{r},n,v} \rangle), \hat{N}_{\vec{r},n,v} = \hat{X}_{\vec{r},n,v}^\dagger \hat{X}_{\vec{r},n,v}, \hat{X}_{\vec{r},n,v}^\dagger = \sum_{\vec{r}',\mu} W_{\vec{r},n,v}(\vec{r}',\mu) \hat{c}_{\vec{r}',\mu}^\dagger$$

$$\text{Unit cell schedule: } \tilde{\mathcal{E}}_{\vec{r},+,B} \circ \tilde{\mathcal{E}}_{\vec{r},+,A} \circ \tilde{\mathcal{E}}_{\vec{r},-,B} \circ \tilde{\mathcal{E}}_{\vec{r},-,A}.$$

What are corresponding update maps on the covariance G ?

$$\text{Define the rank-1 projector: } P_{\vec{r},n,v} = W_{\vec{r},n,v} W_{\vec{r},n,v}^\dagger, Q_{\vec{r},n,v} = 1 - P_{\vec{r},n,v}$$

$$\text{Define the Born-probability: } p_{\vec{r},n,v} = \langle \hat{N}_{\vec{r},n,v} \rangle = W_{\vec{r},n,v}^\dagger C W_{\vec{r},n,v}, G = 2C - 1$$

$$\text{Sherman-Morrison Facts: } \text{Tr}(G P_{\vec{r},n,v}) = 2p_{\vec{r},n,v} - 1$$

$$(-P_{\vec{r},-,v} - G^{-1})^{-1} = -(P_{\vec{r},-,v} + G^{-1})^{-1} = -G + \frac{G P_{\vec{r},-,v} G}{1 + \text{Tr}(G P_{\vec{r},-,v})} = -G + \frac{1}{2p_{\vec{r},-,v}} G P_{\vec{r},-,v} G$$

$$(P_{\vec{r},-,v} - G^{-1})^{-1} = -G - \frac{G P_{\vec{r},-,v} G}{1 - \text{Tr}(G P_{\vec{r},-,v})} = -G - \frac{G P_{\vec{r},-,v} G}{2(1 - p_{\vec{r},-,v})}$$

	Density Matrix	Covariance Matrix
filling outcome 1	$\hat{\rho}' = \frac{\hat{N}_{\vec{r},-,v} \hat{\rho} \hat{N}_{\vec{r},-,v}}{p_{\vec{r},-,v}}$	$G' = P_{\vec{r},-,v} - Q_{\vec{r},-,v} (-P_{\vec{r},-,v} - G^{-1})^{-1} Q_{\vec{r},-,v}$ $= P_{\vec{r},-,v} + Q_{\vec{r},-,v} G Q_{\vec{r},-,v} - \frac{Q_{\vec{r},-,v} G P_{\vec{r},-,v} G Q_{\vec{r},-,v}}{2p_{\vec{r},-,v}}$
filling outcome 2	$\hat{\rho}' = \frac{\hat{X}_{\vec{r},-,v}^\dagger \hat{\rho} \hat{X}_{\vec{r},-,v}}{1 - p_{\vec{r},-,v}}$	$G' = P_{\vec{r},-,v} - Q_{\vec{r},-,v} (P_{\vec{r},-,v} - G^{-1})^{-1} Q_{\vec{r},-,v}$ $= P_{\vec{r},-,v} + Q_{\vec{r},-,v} G Q_{\vec{r},-,v} + \frac{Q_{\vec{r},-,v} G P_{\vec{r},-,v} G Q_{\vec{r},-,v}}{2(1 - p_{\vec{r},-,v})}$

$$\text{Depletion Outcome 1} \quad \dot{e}' = \frac{(1 - \dot{N}_{\bar{r},t,N}) \dot{e} (1 - \dot{N}_{\bar{r},t,N})}{1 - P_{\bar{r},t,N}}$$

$$G' = -P_{\bar{r},t,N} - Q_{\bar{r},t,N} (P_{\bar{r},t,N} - G^{-1})^{-1} Q_{\bar{r},t,N} \\ = -P_{\bar{r},t,N} + Q_{\bar{r},t,N} G Q_{\bar{r},t,N} + \frac{Q_{\bar{r},t,N} G P_{\bar{r},t,N} G Q_{\bar{r},t,N}}{2(1 - P_{\bar{r},t,N})}$$

$$\text{Depletion Outcome 2} \quad \dot{e}' = \frac{\hat{X}_{\bar{r},t,N} \dot{e} \hat{X}_{\bar{r},t,N}^\dagger}{P_{\bar{r},t,N}}$$

$$G' = -P_{\bar{r},t,N} - Q_{\bar{r},t,N} (-P_{\bar{r},t,N} - G^{-1})^{-1} Q_{\bar{r},t,N} \\ = -P_{\bar{r},t,N} + Q_{\bar{r},t,N} G Q_{\bar{r},t,N} - \frac{Q_{\bar{r},t,N} G P_{\bar{r},t,N} G Q_{\bar{r},t,N}}{2P_{\bar{r},t,N}}$$

The overall structure of the matrix update is:

$$G_{t+1} = C_{t+1} - B_{t+1}^\dagger (A_{t+1} - G_t^{-1})^{-1} B_{t+1} = f_{t+1}(G_t)$$

Now suppose we want to study how a small perturbation/mixedness introduced to G relaxes?

$$\text{Initial condition: } \tilde{G}_0 = G_0 + \delta G_0, \quad G_0^2 = 1 \Rightarrow \tilde{G}_0^2 = 1 + \{G_0, \delta G_0\} + \mathcal{O}(\delta^2 G_0) \neq 1 \text{ (impure)}$$

Then $f_{t=1}(\tilde{G}_0) = f_{t=1}(G_0 + \delta G_0)$, Expand to first order in δG

$$f_{t=1}(G_0 + \delta G_0) = C_1 - B_1^\dagger (A_1 - (G_0 + \delta G_0)^{-1})^{-1} B_1$$

$$(G_0 + \delta G_0)^{-1} = [G_0(1 + G_0^{-1} \delta G_0)]^{-1} = (1 + G_0^{-1} \delta G_0)^{-1} G_0^{-1} = G_0^{-1} - G_0^{-1} \delta G_0 G_0^{-1}$$

$$(A_1 - (G_0 + \delta G_0)^{-1})^{-1} = (A_1 - G_0^{-1} + G_0^{-1} \delta G_0 G_0^{-1})^{-1} = (1 + (A_1 - G_0^{-1})^{-1} G_0^{-1} \delta G_0 G_0^{-1})^{-1} (A_1 - G_0^{-1})^{-1}$$

$$= (A_1 - G_0^{-1})^{-1} - (A_1 - G_0^{-1})^{-1} G_0^{-1} \delta G_0 G_0^{-1} (A_1 - G_0^{-1})^{-1}$$

$$= (A_1 - G_0^{-1})^{-1} - (G_0 A_1 - 1)^{-1} \delta G_0 (A_1 G_0 - 1)^{-1}$$

$$f_{t=1}(\tilde{G}_0) = f_{t=1}(G_0) + B_1^\dagger (G_0 A_1 - 1)^{-1} \delta G_0 (A_1 G_0 - 1)^{-1} B_1 = f_{t=1}(G_0) + \delta f_{t=1}(G_0) = G_1 + \delta G_1 (*)$$

Claim: Define $\tilde{G}_t = G_t + \delta G_t$

$$\text{Then } f_{t+1}(\tilde{G}_t) = f_{t+1}(G_t) + \delta f_{t+1}(\delta G_t) (**)$$

$$\text{Where } \delta f_{t+1}(\delta G_t) = \delta G_{t+1} = T_{t+1}^\dagger \delta f_t(\delta G_{t-1}) T_{t+1}, \text{ Where } T_t = (A_t G_{t-1} - 1)^{-1} B_t$$

$$\text{and } f_t(G_{t-1}) = C_t - B_t^\dagger (A_t - G_{t-1}^{-1})^{-1} B_t$$

Proof: We already proved the base case in (*). Suppose (**) holds.

$$\begin{aligned}
 \text{Then } f_{t+2}(\tilde{G}_{t+1}) &= C_{t+2} - B_{t+2}^{\dagger} (A_{t+2} - (G_{t+1} + \delta G_{t+1})^{-1})^{-1} B_{t+2} \\
 &= C_{t+2} - B_{t+2}^{\dagger} (A_{t+2} - G_{t+1})^{-1} B_{t+2} + B_{t+2}^{\dagger} (A_{t+2} G_{t+1} - 1)^{-1} \delta G_{t+1} (G_{t+1} A_{t+2} - 1)^{-1} B_{t+2} \\
 &= f_{t+2}(G_{t+1}) + T_{t+2}^{\dagger} \delta G_{t+1} T_{t+2} = f_{t+2}(G_{t+1}) + \delta f_{t+2}(\delta G_{t+1}) \checkmark
 \end{aligned}$$

Okay, so we should be studying the spectrum of

$$\lim_{n \rightarrow \infty} \frac{1}{2n} \log(\tilde{T}_n \tilde{T}_n^{\dagger}), \quad \tilde{T}_n = T_1 \cdots T_{n-1} T_n, \quad T_t = (A_t G_{t-1} - 1)^{-1} B_t, \quad \text{for } t=1, \dots, n$$

Let's translate this to the adaptive protocol at hand.

Filling outcome 1	$A = -P_{\tilde{r}, -, v} ; B = Q_{\tilde{r}, -, v}, \text{ prob} = p_{\tilde{r}, -, v}$
Filling outcome 2	$A = P_{\tilde{r}, -, v} ; B = Q_{\tilde{r}, -, v}, \text{ prob} = 1 - p_{\tilde{r}, -, v}$
Depletion outcome 1	$A = P_{\tilde{r}, +, v} ; B = Q_{\tilde{r}, +, v}, \text{ prob} = 1 - p_{\tilde{r}, +, v}$
Depletion outcome 2	$A = -P_{\tilde{r}, +, v} ; B = Q_{\tilde{r}, +, v}, \text{ prob} = p_{\tilde{r}, +, v}$

Thus, the transfer matrix associated with an elementary stochastic process is

$$T_t = \begin{cases} -(P_{\tilde{r}, n, v} G_{t-1} + 1)^{-1} Q_{\tilde{r}, n, v}, & \text{prob} = p_{\tilde{r}, n, v} \\ (P_{\tilde{r}, n, v} G_{t-1} - 1)^{-1} Q_{\tilde{r}, n, v}, & \text{prob} = 1 - p_{\tilde{r}, n, v} \end{cases}$$

Since $P_{\tilde{r}, n, v}$ is a rank-1 projector, we can simplify it further via Sherman-Morrison.

Using $-(PG + 1)^{-1} = -1 + \frac{1}{2p} PG ; (PG - 1)^{-1} = -1 - \frac{1}{2(1-p)} PG$

$$T_t = -Q_{\tilde{r}, n, v} + \begin{cases} \frac{P_{\tilde{r}, n, v} G_{t-1} Q_{\tilde{r}, n, v}}{2p_{\tilde{r}, n, v}}, & \text{prob} = p_{\tilde{r}, n, v} \\ -\frac{P_{\tilde{r}, n, v} G_{t-1} Q_{\tilde{r}, n, v}}{2(1-p_{\tilde{r}, n, v})}, & \text{prob} = 1 - p_{\tilde{r}, n, v} \end{cases}$$